

7 Baseline MAH Partial Equilibrium

Fix the institutional regime $M \in \{0, 1\}$, the developer distribution $H(a, k)$, the project distribution $F(q, m)$, the qualified-supplier distribution $H_C(z)$, and all other exogenous primitives introduced above. A baseline MAH partial equilibrium is the collection

$$\boxed{\{p_m^*, x_i^*, r_i^*(q, m), s_j^*\}_{i,j}}. \quad (1)$$

The suppressed arguments are $x_i^* = x_i^*(M, p_m^*)$, $r_i^*(q, m) = r_i^*(q, m; M, p_m^*)$, and $s_j^* = s_j^*(p_m^*, z_j)$. The collection contains exactly these four families of endogenous optimal objects.

It satisfies the following four conditions.

1. Optimal commercialization route. For every planning-stage project–developer pair outside the zero-measure tie set, use the route-indexed shorthand W_i^r for the four Phase 4 values, with $W_i^T = W^T$ and $W_i^A = W^A$. Then

$$r_i^*(q, m; M, p_m^*) \in \arg \max_{r \in \{I, E, T, A\}} W_i^r(q, m; M, p_m^*), \quad (2)$$

with route E unavailable when $M = 0$. This is the deterministic route choice defined above; route E retains holder rights and is not route T .

2. Optimal project advancement. For every developer i , the common project-advancement control is

$$x_i^*(M, p_m^*) = \left[\frac{\beta a_i \Omega_i(M, p_m^*)}{\kappa} \right]^{1/\nu}, \quad (3)$$

including the zero-value corner $x_i^* = 0$. There is no class-specific x_{ig} control.

3. Optimal qualified-supplier capacity. For every qualified supplier j ,

$$s_j^*(p_m^*, z_j) \in \arg \max_{s_j \geq 0} \{p_m^* s_j - \Psi(s_j; z_j)\}. \quad (4)$$

Strict convexity of Ψ makes this capacity choice unique. Supplier entry is not a margin of the model.

4. Qualified-capacity market clearing. The price p_m^* is the unique solution to

$$D_m(p_m^*; M) = S_m(p_m^*). \quad (5)$$

The qualified-capacity existence and uniqueness conditions apply. The demand and supply functions determine the price but are not additional members of the equilibrium collection.

7.1 Closure boundary

Equations (2)–(5) close only commercialization route choice, common project advancement, qualified-supplier capacity, and the manufacturing-service price. There is no entry condition and no labor-market, capital-market, product-market aggregate, firm-distribution, or welfare-clearing condition. Product price $p^*(c)$ remains a conditional firm-level optimum from the pricing problem, not a second market-clearing price.